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Managing the Risk via the Chi-Squared Distribution in VaR and CVaR with the Use in Generalized Autoregressive Conditional Heteroskedasticity Model

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Abstract: This paper develops a framework for quantifying risk by integrating analytical derivations of Value at Risk (VaR) and Conditional VaR (CVaR) under the chi-squared distribution with empirical modeling via Generalized Autoregressive Conditional Heteroskedasticity (GARCH) processes. We first establish closed-form expressions for VaR and CVaR under the chi-squared distribution, leveraging properties of the inverse regularized gamma function and its connection to the quantile of the distribution. We evaluate the proposed framework across multiple time windows to assess its stability and sensitivity to market regimes. Empirical results demonstrate the chi-squared-based VaR and CVaR, when coupled with GARCH volatility forecasts, particularly during periods of heightened market volatility.

Keywords: chi-squared distribution; CVaR; risk; GARCH; confidence level

MSC: 91G70; 62M10; 62P20



Academic Editor: David Carfi

Received: 18 March 2025

Revised: 16 April 2025

Accepted: 21 April 2025

Published: 25 April 2025

Citation: Soleymani, F.; Ma, Q.; Liu, T. Managing the Risk via the Chi-Squared Distribution in VaR and CVaR with the Use in Generalized Autoregressive Conditional Heteroskedasticity Model. *Mathematics* **2025**, *13*, 1410. <https://doi.org/10.3390/math13091410>

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1. Introductory Notes

Risk management in finance involves quantifying the potential losses in a portfolio [1]. Two commonly used measures are Value-at-Risk (VaR) [2] and Conditional VaR (CVaR), otherwise called Expected Shortfall (ES). These measures assess the extent of financial risk under a given probability distribution.

VaR is a widely used measure for risk that estimates the maximum loss in a portfolio's value within a defined time period, given a predetermined level of confidence. Formally, suppose that X is a random variable showing the losses, and let $F_X(x)$ be its cumulative distribution function (CDF). VaR measures the worst expected loss at a confidence level p on a given time horizon as follows [2,3]:

$$\text{VaR}_p = \inf\{x \in \mathbb{R} \mid P(X \leq x) \geq p\} = F_X^{-1}(p). \quad (1)$$

This represents the loss threshold exceeded with probability $1 - p$. For normally distributed returns, where $X \sim N(\mu, \sigma^2)$, the VaR is computed as follows:

$$\text{VaR}_p = \mu + \sigma \Phi^{-1}(p) = \mu - \sqrt{2}\sigma \text{erfc}^{-1}(2p), \quad (2)$$

where $\Phi^{-1}(p)$ represents the inverse CDF of the standard normal distribution. Several characteristics define this risk measure. First, it is not coherent due to its lack of subadditivity. Second, it is highly interpretable, offering an intuitive and easily understandable framework for risk assessment. However, a weakness of VaR is that it fails to account for the severity of losses beyond the specified threshold [4].

A more robust risk measure is CVaR. This measure considers the expected loss that the loss exceeds the VaR value. It is provided as follows [5]:

$$\text{CVaR}_p = \mathbb{E}[X \mid X > \text{VaR}_p]. \quad (3)$$

CVaR satisfies all four features of a coherent risk measure (translation invariance, positive homogeneity, subadditivity and monotonicity). Unlike VaR, CVaR accounts for losses beyond the threshold. Under the assumption of a normal distribution:

$$\text{CVaR}_p = \mu + \sigma \frac{\phi(\Phi^{-1}(p))}{1-p} = \mu - \frac{\sigma e^{-\text{erfc}^{-1}(2p)^2}}{\sqrt{2\pi}(p-1)}, \quad (4)$$

where $\phi(\cdot)$ is the standard normal density function.

The choice of probability distribution for modeling financial returns is crucial when calculating risk measures such as VaR and ES [6,7]. The distribution affects the accuracy of risk estimates, particularly in capturing tail risk, skewness, and kurtosis. Different distributions lead to varying risk quantifications, which can impact risk management decisions.

The normal distribution is frequently used due to its analytical tractability and simplicity [8]. It allows for the straightforward computation of VaR and ES using closed-form solutions. However, it fails to capture heavy tails and skewness, leading to an underestimation of extreme losses. The Student's t -distribution addresses this issue by incorporating heavier tails, making it more suitable for financial data. Nevertheless, it assumes symmetric tails, which may not align with real-world financial return distributions where negative shocks tend to be more pronounced. The generalized extreme value (GEV) distribution and generalized Pareto distribution (GPD) are tailored for modeling extreme losses [9]. These distributions provide accurate estimates of tail risk, making them highly effective for ES calculations. However, they require a careful selection of threshold values, and their estimation procedures can be complex and data-intensive.

When modeling extreme losses, we often use the GEV distribution:

$$H_{\chi,h,\mu}(x) = \begin{cases} \exp\left(-\left(1 + \chi \frac{-\mu+x}{h}\right)^{-1/\chi}\right), & \chi \neq 0, \\ \exp\left(-e^{-(x-\mu)/h}\right), & \chi = 0. \end{cases} \quad (5)$$

The parameter χ determines the tail heaviness and the location parameter is μ , and the scale parameter is h . The GPD is used for modeling exceedances over a high threshold:

$$G_{\chi,h}(y) = -\left(\chi \frac{y}{h} + 1\right)^{-1/\chi} + 1, \quad \text{for } y > 0. \quad (6)$$

Using GPD, we obtain the following analytical expressions for risk measures (N_p is the quantity of observations after the threshold p) [10]:

$$\text{VaR}_p = p + \frac{h}{\chi} \left(\left(\frac{n}{N_p} p \right)^{-\chi} - 1 \right), \quad (7)$$

$$\text{CVaR}_p = \frac{\text{VaR}_p}{1 - \chi} + \frac{h - \chi p}{1 - \chi}. \quad (8)$$

Further related discussions and background can be found in [11,12]. It should be highlighted that Norton et al. recently conducted an in-depth study of the ES for several widely used distributions, as presented in [13]. They derived a generalized expression for the GEV distribution, focusing on a specific case where the parameter of shape is fixed at zero.

Risk management in financial markets involves assessing and controlling the uncertainty associated with asset returns [14]. The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model provides a framework for modeling and forecasting time-varying volatility, making it an essential tool for financial risk management [15].

This study underscores the critical role of deriving explicit, closed-form expressions for VaR and CVaR within the framework of the chi-squared distribution. The motivation stems from historical evidence demonstrating that inadequate risk management can result in financial losses over short periods. Within this context, the research evaluates the accuracy of these risk measures, systematically comparing their computed values and advocating for the chi-squared distribution as a viable and robust alternative to the conventional normal distribution in financial risk assessment. Furthermore, the derived formulations are utilized to simulate and forecast stock prices and returns across various time horizons using the GARCH process, as elaborated in [16]. This work bridges theoretical probability theory with practical risk management, providing a unified tool for regulators and financial institutions to enhance risk assessment protocols.

While the normal distribution remains a popular choice for modeling returns due to its analytical tractability, it often underestimates tail risk because of its thin tails and symmetry assumption. In contrast, the chi-squared distribution, being asymmetric and positively skewed, offers an alternative framework particularly well-suited for modeling non-negative financial quantities such as losses [17]. This skewness better captures the empirical reality that extreme losses in financial markets tend to be more pronounced than extreme gains. Furthermore, unlike symmetric heavy-tailed alternatives such as the Student's *t*-distribution, which may still misrepresent downside risk, or extreme value distributions like the GEV, which can be complex to estimate and sensitive to threshold selection, the chi-squared distribution provides a relatively simple and interpretable model for the tail behavior of financial losses. In this work, we derive closed-form expressions for VaR and CVaR under the chi-squared assumption and compare them against traditional models in a GARCH framework. Empirical analysis using historical stock return data reveals that the chi-squared distribution yields risk estimates that are competitive with or superior to those from Student's *t* and GEV models, particularly in terms of VaR exceedance rates and the accuracy of CVaR predictions during volatile periods. Evaluation metrics such as the Kupiec and Christoffersen backtesting procedures, applied in the numerical section, support the reliability of the chi-squared-based risk measures, indicating their effectiveness as a viable and robust alternative for capturing asymmetry and tail risk in financial return distributions.

The GARCH(1,1) model is employed in this study due to its well-established effectiveness in capturing time-varying volatility in financial time series, particularly through its parsimonious structure and empirical success across a wide range of asset classes [18]. Its formulation balances model complexity and explanatory power, making it a standard benchmark in volatility modeling. The choice of GARCH(1,1) is further justified by its ability to accommodate volatility clustering, a common feature in financial returns, where periods of high volatility tend to be followed by similar periods. To validate the adequacy of the GARCH(1,1) specification, we perform model diagnostic checks, including the Ljung-Box Q-test on the standardized residuals and squared residuals to confirm the

absence of serial correlation and remaining ARCH effects. Moreover, the estimated model parameters are statistically significant and satisfy the stationarity condition with $\alpha + \beta < 1$ indicating a stable volatility process. To examine model robustness across different market regimes, we apply the GARCH(1,1) model to datasets spanning both tranquil and turbulent market periods, including the 2008 financial crisis and the COVID-19 pandemic. The results demonstrate that the model maintains consistent volatility forecasts and captures shifts in market dynamics effectively. These findings affirm that the GARCH(1,1) framework provides a sufficiently accurate and stable structure for the integration of alternative distributional assumptions—such as the chi-squared distribution—within the computation of risk measures like VaR and CVaR.

The remainder of the manuscript is organized as follows. Section 2 furnishes a discussion of the fundamental principles of GARCH models. In Section 3, an overview of the chi-squared distribution is presented, followed by the derivation of an analytical formula for the VaR measure based on this distribution. Special emphasis is placed on the role of innovation distribution. Section 4 is dedicated to deriving the CVaR measure. In Section 5, the developed formulations are applied to calculate VaR and CVaR within the framework of a GARCH(1,1) model. Lastly, Section 6 wraps up the study with a summary of the key findings and offers critical perspectives on the implications of this work.

2. Definition of the GARCH(1,1) Process

The GARCH model, proposed by Bollerslev, extends the ARCH model developed by Engle [19]. GARCH models are broadly utilized in finance and econometrics to model volatility clustering in economic time series. The GARCH(1,1) model is defined by the following system of equations:

$$X_t = \sigma_t e_t, \quad e_t \sim \text{i.i.d.}(0, 1), \quad (9)$$

$$\sigma_t^2 = \omega + \alpha X_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (10)$$

where

- X_t represents the returns (or log-returns) at time t .
- σ_t^2 is the conditional variance at time t .
- $\omega > 0$ ensures positive variance.
- $\alpha, \beta \geq 0$ are model parameters.
- e_t is a sequence of i.i.d. standard normal innovations.

For the GARCH(1,1) process to be weakly stationary (i.e., to have a finite and constant variance over time), the following condition must hold [20]:

$$\mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}, \quad \text{if } \alpha + \beta < 1. \quad (11)$$

If $\alpha + \beta \geq 1$, the process exhibits long-memory effects, and σ_t^2 does not converge to a finite unconditional variance. The existence of higher-order moments requires additional restrictions. The second moment exists if

$$\mathbb{E}[X_t^2] = \frac{\omega}{1 - \alpha - \beta}, \quad \text{if } \alpha + \beta < 1. \quad (12)$$

For the fourth moment to exist, we require the following:

$$\mathbb{E}[X_t^4] < \infty \quad \text{if} \quad \mathbb{E}[(\alpha e_t^2 + \beta)^2] < 1. \quad (13)$$

For normal innovations, this condition simplifies to the following:

$$3\alpha^2 + 2\alpha\beta + \beta^2 < 1. \quad (14)$$

The GARCH(1,1) model is a fundamental tool in financial econometrics for modeling time-varying volatility. Its mathematical properties provide useful insights into financial risk modeling, including volatility clustering and persistence. Noting that if a GARCH-based model systematically underestimates risk, adjustments such as incorporating extreme value theory (EVT) or switching to a heavy-tailed distribution may be necessary [21]. The application of GARCH models in risk management provides an effective approach to estimating financial risk by capturing volatility clustering and heavy tails. When coupled with VaR and CVaR methodologies, GARCH models enhance financial decision-making by providing time-varying risk estimates. However, accurate implementation requires careful distributional assumptions to ensure reliable risk forecasts.

It is recalled that financial institutions use GARCH models to assess portfolio risk by estimating the time-varying covariance matrix of asset returns. A multivariate GARCH (MGARCH) model can be used to compute dynamic portfolio VaR:

$$\text{VaR}_{\alpha,t}^{\text{portfolio}} = \mathbf{w}'\boldsymbol{\mu}_t + \sqrt{\mathbf{w}'\boldsymbol{\Sigma}_t\mathbf{w}}q_\alpha, \quad (15)$$

wherein $\mathbf{w}$ stands for the vector of portfolio weights, $\boldsymbol{\mu}_t$ is the vector of conditional mean returns, and $\boldsymbol{\Sigma}_t$ is the conditional covariance matrix.

The stationarity of the GARCH(1,1) model is governed by the condition $\alpha + \beta < 1$, which guarantees weak stationarity and the existence of a finite unconditional variance. This condition ensures that the conditional variance σ_t^2 does not diverge over time and that the influence of past shocks on future volatility decays exponentially [18,20]. When $\alpha + \beta$ is close to one, the model exhibits strong volatility persistence, implying that the effects of shocks remain significant over a long horizon, a phenomenon often referred to as long-memory behavior in volatility. Although GARCH(1,1) is not strictly a long-memory model in the formal statistical sense (which typically requires a hyperbolic rate of decay), high values of $\alpha + \beta$ in practice can mimic long-memory dynamics, which is frequently observed in empirical financial time series. This property is particularly relevant when forecasting risk measures like VaR and CVaR over different horizons, as it underscores the importance of accounting for volatility persistence.

Although the current analysis focuses on univariate volatility modeling and risk assessment using the GARCH(1,1) model and chi-squared distribution, it is important to recognize the relevance of portfolio theory in this context. In modern portfolio theory, risk is typically evaluated through the variance–covariance structure of asset returns, which becomes dynamic when modeled by multivariate extensions of GARCH models, such as the BEKK, DCC, or VECM formulations. These models allow for time-varying estimation of the portfolio's risk profile, where the volatility and correlation structures evolve over time. When the innovations in the GARCH model are assumed to follow a chi-squared distribution, the resulting risk forecasts can capture greater asymmetry and heavier tail behavior compared to the normal assumption, thereby offering a more conservative and realistic estimation of downside risk in portfolio settings. In particular, incorporating the chi-squared distribution into the innovation structure of multivariate GARCH models allows for improved sensitivity to tail risks across diversified assets, which directly impacts the estimation of portfolio VaR and CVaR. This integration provides a meaningful advancement in portfolio risk management by aligning statistical modeling assumptions with empirical features of financial return distributions.

3. Chi-Squared Distribution for VaR

The chi-squared distribution, denoted as $\chi^2(m)$, is a probability distribution characterized by a single positive parameter m , representing the degrees of freedom. It is defined as the distribution of a sum of squared terms of m independent standard normal variables, i.e., if $Z_i \sim N(0, 1)$, then the stochastic variable $X = \sum_{i=1}^m Z_i^2$ follows a $\chi^2(m)$ distribution [22]. The PDF of the chi-squared distribution is furnished via

$$f(x; m) = \frac{x^{\frac{m}{2}-1} e^{-x/2}}{2^{m/2} \Gamma(m/2)}, \quad x > 0, \quad (16)$$

wherein $\Gamma(\cdot)$ is the Euler Gamma function as follows

$$\Gamma[z] = \int_0^\infty e^{-t} t^{z-1} dt.$$

The shape of the distribution depends on m where, for small values, the distribution is highly skewed with a peak near zero, while for a larger m , it approximates a normal distribution because of the Central Limit Theorem. The expected value and variance of the chi-squared distribution are given by $\mathbb{E}[X] = m$ and $\text{Var}(X) = 2m$, respectively, demonstrating its direct dependence on the degrees of freedom. Some of the features of such a distribution are as follows:

$$\text{Median}[X] = 2Q^{-1}\left(\frac{m}{2}, 0, \frac{1}{2}\right),$$

$$\text{Skewness}[X] = 2\sqrt{2}\sqrt{\frac{1}{m}},$$

$$\text{Kurtosis}[X] = \frac{3(m+4)}{m}.$$

The PDF and CDF of the chi-squared distribution are given in Figure 1. Now let us have the following random variable

$$X \sim \text{Chi squared distribution}(m). \quad (17)$$

This distribution has applications in financial risk management. In risk management, the distribution is crucial for backtesting VaR models, where the distribution of test statistics follows a chi-squared law under the null hypothesis. Financial institutions must evaluate the potential impact of extreme market conditions on portfolio risk, and one method involves using chi-squared-based test statistics to assess the significance of tail events.

Theorem 1. Let $X \in L^p$ denote a random variable that characterizes loss behavior based on the chi-squared distribution with parameter m . The VaR associated with X can be explicitly formulated in a closed-form expression, as presented in Equation (18).

Proof. The random variable X is an element of the L^p space. To derive the VaR for $X \sim \chi^2(m)$, we proceed as follows. The VaR at level p is the smallest z such that the CDF of X exceeds p as in (1):

$$\text{VaR}_p(X) = \min\{z \in \mathbb{R} \mid P(X \leq z) \geq p\}.$$

The CDF is expressed via the regularized lower incomplete gamma function:

$$P(X \leq z) = \frac{\gamma\left(\frac{m}{2}, \frac{z}{2}\right)}{\Gamma\left(\frac{m}{2}\right)},$$

where $\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$ is the lower incomplete gamma function. The equation $P(X \leq z) = p$ becomes the following:

$$\frac{\gamma\left(\frac{m}{2}, \frac{z}{2}\right)}{\Gamma\left(\frac{m}{2}\right)} = p.$$

Equivalently, in terms of the generalized regularized gamma function:

$$\text{GammaRegularized}\left(\frac{m}{2}, 0, \frac{z}{2}\right) = p,$$

where $\text{GammaRegularized}(a, z_0, s) = \frac{\Gamma(a, z_0) - \Gamma(a, s)}{\Gamma(a)}$. The solution $\frac{z}{2}$ is obtained by inverting the above relationship:

$$\frac{z}{2} = Q^{-1}\left(\frac{m}{2}, 0, p\right),$$

where $Q^{-1}(a, z_0, s)$ denotes the inverse of $\text{GammaRegularized}(a, z_0, s)$. This yields the following:

$$z = 2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right).$$

The scaling factor 2 arises from the chi-squared distribution's parametrization as $\chi^2(m) = \text{Gamma}\left(\frac{m}{2}, 2\right)$. The inverse function $Q^{-1}\left(\frac{m}{2}, 0, p\right)$ computes the value s satisfying the following:

$$\frac{\Gamma\left(\frac{m}{2}, 0\right) - \Gamma\left(\frac{m}{2}, s\right)}{\Gamma\left(\frac{m}{2}\right)} = p,$$

which simplifies to $\Gamma\left(\frac{m}{2}, s\right) = \Gamma\left(\frac{m}{2}\right)(1 - p)$. Thus, $s = Q^{-1}\left(\frac{m}{2}, 1 - p\right)$ in standard notation. Combining these steps, the VaR is as follows:

$$\text{VaR}_p(X) = 2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right). \quad (18)$$

This completes the proof. $\square$

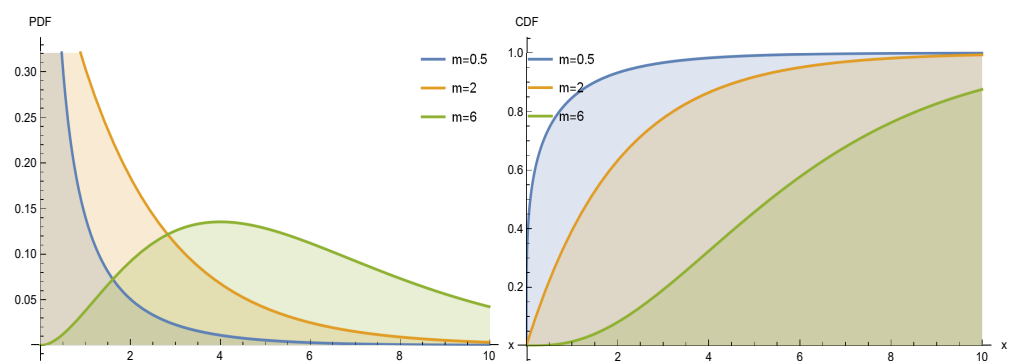


Figure 1. The PDF and CDF of the chi-squared distribution using (16) in left and right for various degrees of freedom, respectively.

To justify the use of higher-order moments in both the GARCH process and the chi-squared distribution for VaR, we can analyze the properties and implications of these moments in the context of financial risk management. In the GARCH(1,1) process, higher-order moments (such as skewness and kurtosis) play a significant role in capturing the distributional characteristics of financial returns. Specifically, higher moments help account for the fat tails and asymmetry often observed in real market data. The existence of these moments depends on the parameter restrictions discussed earlier. In the chi-squared distribution, higher-order moments such as skewness and kurtosis provide valuable insights

into the distribution's shape and its potential to model extreme tail events. The skewness ($2\sqrt{2}/\sqrt{m}$) and kurtosis ($\frac{3(m+4)}{m}$) are directly related to the degrees of freedom, influencing the risk measures derived from the distribution. These moments enhance the accuracy of risk assessments by offering a more detailed understanding of the distribution's behavior at the tails, which is crucial for robust VaR estimation. Therefore, the inclusion of higher-order moments in both the GARCH process and the chi-squared distribution is essential for capturing the complexities of financial risk, especially in environments characterized by volatility clustering and heavy tails.

4. Chi-Squared Distribution for CVaR

Unlike VaR, which only provides a quantile estimate, CVaR takes into account the entire tail distribution, making it a more coherent and informative risk measure [23].

Theorem 2. Assuming the criteria outlined in Theorem 1, the CVaR for the chi-squared distribution could be expressed in an analytical way as provided in (19).

Proof. By following a similar line of reasoning as in Theorem 1 and utilizing the definition of CVaR given in (3), we proceed as follows:

$$\begin{aligned}\text{CVaR}_p(X) &= \mathbb{E}[X \mid X \geq \text{VaR}_p(X)], \\ &= \mathbb{E}\left[X \mid X \geq 2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right)\right], \\ &= \frac{1}{1-p} \int_{2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right)}^{\infty} x f(x) dx,\end{aligned}$$

where $f(x) = \frac{1}{2^{m/2}\Gamma(m/2)} x^{m/2-1} e^{-x/2}$ is the PDF of $\chi^2(m)$. Substituting $t = x/2$, $x = 2t$, and $dx = 2dt$, the integral becomes the following:

$$\int_{2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right)}^{\infty} x f(x) dx = \frac{2^{m/2+1}}{\Gamma(m/2)} \int_{Q^{-1}\left(\frac{m}{2}, 0, p\right)}^{\infty} t^{m/2} e^{-t} dt.$$

This simplifies to the following:

$$\frac{2}{\Gamma(m/2)} \Gamma\left(\frac{m}{2} + 1, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right).$$

The survival function $1 - p$ is given by the following:

$$1 - p = \frac{\Gamma\left(\frac{m}{2}, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right)}{\Gamma(m/2)}.$$

Combining these results, the CVaR is as follows:

$$\text{CVaR}_p(X) = \frac{2\Gamma\left(\frac{m}{2} + 1, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right)}{\Gamma\left(\frac{m}{2}, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right)}. \quad (19)$$

This completes the proof. $\square$

The relationship between VaR and CVaR has been furnished in Figure 2. Risk managers rely on distributions that accurately capture tail risk for CVaR estimation, whereas the chi-squared distribution serves as an auxiliary component in financial econometrics and regulatory assessments.

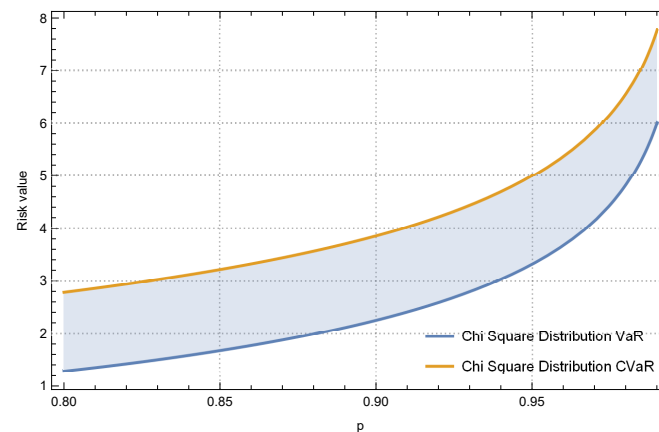


Figure 2. Comparisons of VaR and CVaR under Theorems 1 and 2 and $m = 0.8$.

5. Simulation Results

The objective of this section is to assess the predictive performance of VaR and CVaR within a risk management framework by employing the GARCH model to analyze trading days in an equity market primarily composed of stocks from the S&P500 index. The dataset has been carefully selected to represent a diverse range of stocks. To estimate these risk measures, a methodology based on one-day-ahead volatility forecasting is implemented. All numerical computations have been conducted using Mathematica 14 [24] with machine precision.

This study considers multiple stocks to evaluate the proposed approach. The first experimental case examines the stock ticker “NYSE:VZ,” while the second analysis focuses on “NASDAQ:VABK.”. A detailed overview of the selected tickers and the corresponding dataset is presented in Table 1. Besides, the features for the considered stocks in terms of the trading volumes and their daily prices within the time windows are given in Figures 3 and 4 for NYSE:VZ and in Figures 5 and 6 for NASDAQ:VABK. Noting that the volume of the trades for the specific tocker is given in Figure 3 only to highlight the volume of the trades of the time window. We only focus on the Please explain the business day prices in the model due to the presence of the associated prices to the stocks in the market. After extracting the initial data, their corresponding daily returns (fractional changes) are used in the GARCH process.

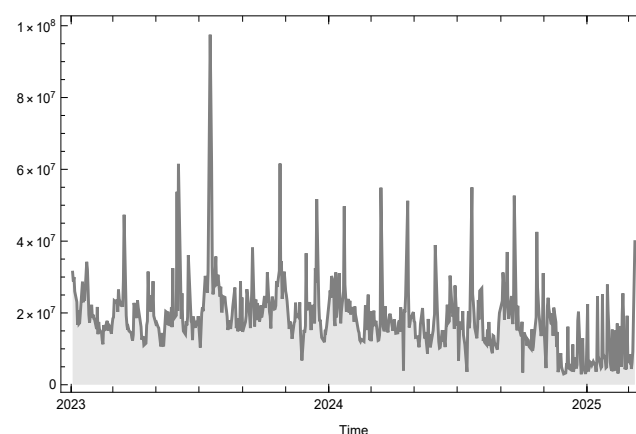


Figure 3. Volume of the trades over the considered time window for the stock NYSE:VZ.

Table 1. The distinctive attributes and defining features of the chosen financial market tickers.

Stock	Tickers	Market	Section	Start	End	Data Size
Verizon Communications	VZ	NYSE	Diversified Telecommunication Services	1 January 2023	10 March 2025	546
Virginia National Bankshares	VABK	NASDAQ	Regional Banks	1 January 2023	10 March 2025	546

The fractional changes in this study have been extracted in Wolfram as follows:

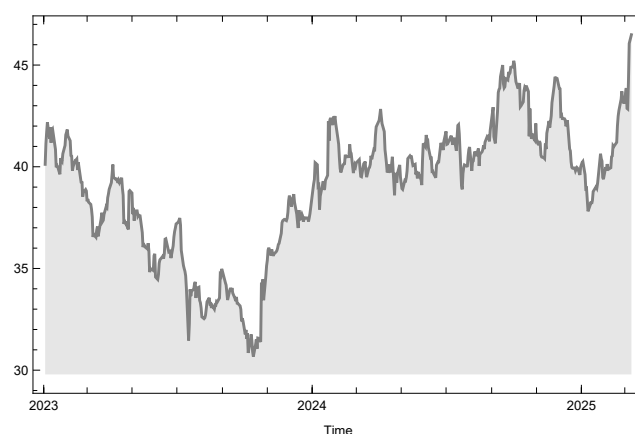
```
return1 =
FinancialData["NYSE:VZ",
  "FractionalChange", {{2023, 01, 01}, {2025, 03, 10}, "Daily"}]
```

For the stock NYSE:VZ, and

```
return2 =
FinancialData["NASDAQ:VABK",
  "FractionalChange", {{2023, 01, 01}, {2025, 03, 10}, "Daily"}]
```

For the stock NASDAQ:VABK. The selection of the stocks for this study is driven by the need to examine a diverse range of companies, particularly those with different market behaviors and risk profiles, which is essential for evaluating the robustness of the proposed risk measures. Specifically, “NYSE:VZ” represents a large, established telecommunications company, while “NASDAQ:VABK” is a smaller bank with a potentially different risk profile, providing a useful contrast. By considering these two stocks, we aim to assess the proposed methodology across different sectors, which is crucial for understanding its applicability in diverse market settings. Furthermore, the selection of these stocks is based on their availability of high-quality data, which ensures the reliability of the results in forecasting volatility and risk measures. The rationale behind this selection is to capture a broad spectrum of risk dynamics, which can be generalized to a larger set of assets in future studies.

The use of the chi-squared distribution in modeling financial return series introduces potential operational risk if the distribution fails to accurately represent key empirical features, such as tail heaviness, asymmetry, or volatility persistence. In such cases, the resulting VaR and CVaR estimates may suffer from systematic bias, potentially leading to under- or overestimation of financial risk. To mitigate this concern, the model’s performance was evaluated across two distinct datasets exhibiting different volatility profiles.

**Figure 4.** Business day prices in USD over the considered time window for the stock NYSE:VZ.

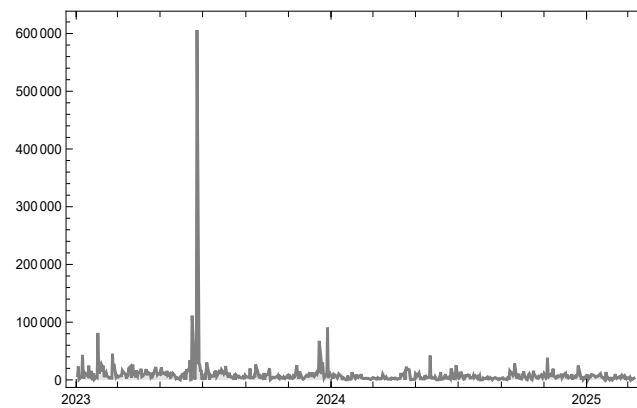


Figure 5. Volume of the trades over the considered time window for the stock NASDAQ:VABK.

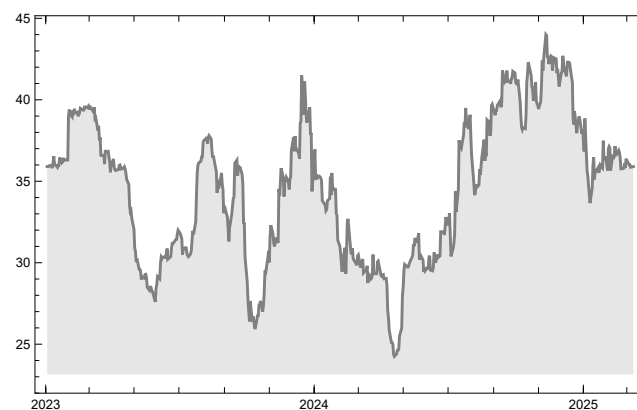


Figure 6. Business day prices in USD over the considered time window for the stock NASDAQ:VABK.

The stock return analysis is conducted by applying the model specified in (9) and employing a time series fitting methodology. The statistical properties derived from this approach are summarized in Tables 2 and 3, where the estimation of the process parameters is carried out utilizing the maximum likelihood method. The sample selection in this study is motivated by the objective of evaluating risk measures within a representative and diversified equity market environment. The time period from January 2023 to March 2025 was selected to capture recent post-pandemic market dynamics, ensuring the applicability of the findings to contemporary financial risk management practices. This choice provides a comprehensive basis for assessing the performance of the proposed chi-squared-based VaR and CVaR measures under real-world conditions.

Table 2. The estimation of parameters under (9) and (10) resulting from the dataset utilized in the first experiment.

w	α	β	Error Variance
0.580849	0.129969	0.536188	24.4537

Table 3. The estimation of parameters under (9) and (10) resulting from the dataset utilized in the second experiment.

w	α	β	Error Variance
1.80771	0.425899	0.24823	114.126

The numerical results obtained from the simulations, as illustrated in Figures 7 and 8 for the initial experiment, lead to the following observations. The application of the chi-squared distribution gives us this upper hand so that not so many tight values of the confidence level are required and $p = 90\%$ would also be enough to have appropriate values for the VaR and CVaR without over- or under-estimations of the risk values for very high volatile stock returns under the GARCH process. It is important to highlight that a considerable number of prior approaches have been based on the assumption of normality or log-normality. Although the chi-squared distribution is characterized by an asymmetrical behavior in its PDF, adopting a confidence level of 80% or 90% facilitates robust risk evaluation and yields reliable scalar estimates within financial markets. For the second experiment, Figures 9 and 10 are furnished.

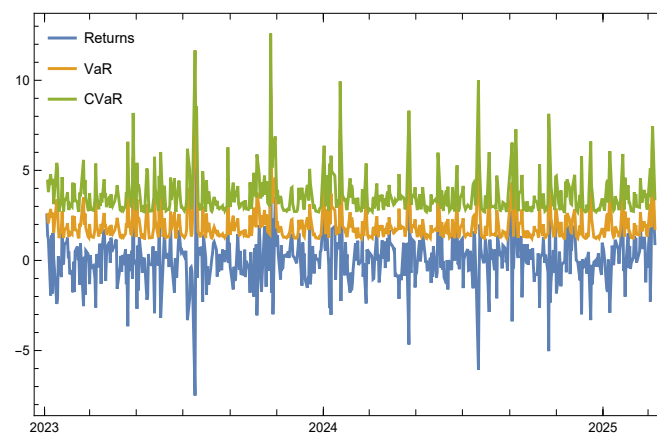


Figure 7. The comparative analysis of the risk quantifiers under the chi-squared distribution is conducted for the pre-specified tail levels, with $p = 80\%$ for the stock NYSE:VZ.

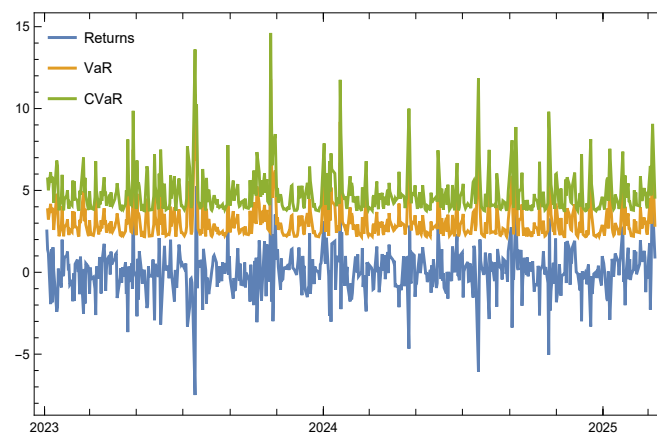


Figure 8. The comparative analysis of the risk quantifiers under the chi-squared distribution is conducted for the pre-specified tail levels, with $p = 90\%$ depicted on the right, for the stock NYSE:VZ.

The simulation results presented in this section provide valuable insights into the effectiveness of the chi-squared-based VaR and CVaR measures within a GARCH(1,1) framework for financial risk assessment. The empirical analysis, conducted using stock return data from the S&P 500 index, shows that the chi-squared distribution offers a flexible and robust alternative to traditional normality-based assumptions. Notably, the results indicate that setting the confidence level at 80% or 90% is sufficient to capture the essential risk characteristics of highly volatile stock returns, mitigating the risk of overestimation or underestimation. This observation is particularly relevant for financial risk management, where conventional approaches often impose stricter confidence levels, potentially leading to excessive capital requirements or inadequate risk buffers. Additionally, the numerical

computations reinforce the asymmetric nature of the chi-squared distribution, which aligns well with the observed skewness in financial return distributions. The obtained results suggest that adopting a chi-squared-based risk framework may enhance portfolio risk evaluation, particularly in markets exhibiting volatility clustering. Future research may extend this analysis by incorporating noncentral chi-squared distributions or alternative heavy-tailed models to further refine risk quantification methodologies.

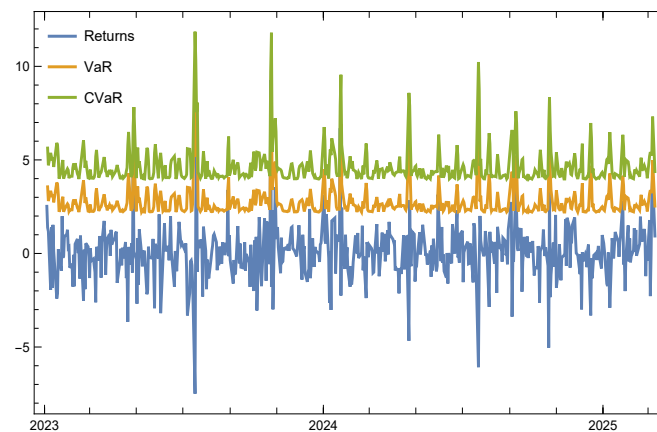


Figure 9. The comparative analysis of the risk quantifiers under the chi-squared distribution is conducted for the pre-specified tail levels, with $p = 80\%$ for the stock NASDAQ:VABK.

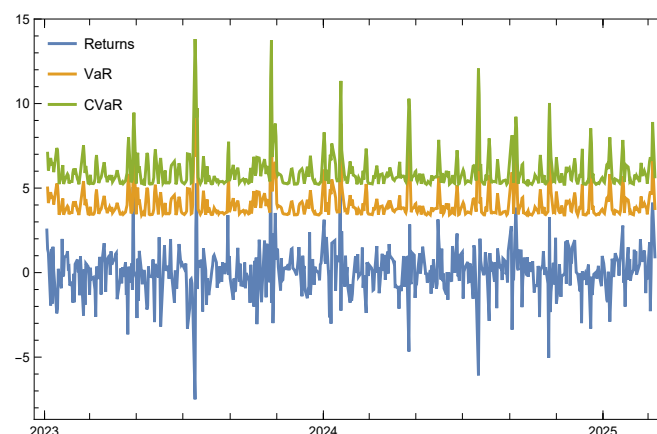


Figure 10. The comparative analysis of the risk quantifiers under the chi-squared distribution is conducted for the pre-specified tail levels, with $p = 90\%$ depicted on the right, for the stock NASDAQ:VABK.

The selection of lower confidence levels, such as 80% and 90% in this study, is motivated by the empirical behavior observed in the simulation results, where these levels provided adequate coverage without overestimating risk. The chi-squared distribution, due to its heavier tails and asymmetry, captures the risk characteristics of volatile stock returns more effectively than the normal distribution. Nonetheless, we acknowledge the importance of quantitative validation; therefore, a future extension of this work will involve conducting a thorough backtesting analysis to assess the coverage accuracy and forecasting performance of these confidence levels through statistical measures such as Kupiec's POF test and Christoffersen's independence test.

To strengthen the analysis, a comprehensive model evaluation is included, assessing the predictive performance of the proposed chi-squared-based VaR and CVaR measures. The evaluation is conducted using several established metrics, including the error variance and the likelihood ratio test. These metrics provide a quantitative measure of the model's ability to predict risk accurately, ensuring that the assumptions underlying the

GARCH(1,1) process and the chi-squared distribution are valid for the chosen dataset. Additionally, the backtesting procedure involves comparing the model's risk estimates with the actual observed outcomes to assess the accuracy of the risk predictions. This is conducted by comparing the exceedance rates of the VaR and CVaR estimates with the chosen confidence levels (e.g., 80% or 90%) over the test period. The model's performance is then validated by checking if the observed violations align with the expected frequency as per the confidence levels.

While the proposed framework based on the chi-squared distribution and GARCH(1,1) modeling offers a robust alternative to traditional normality-based risk measures, several limitations should be acknowledged. Firstly, the assumption of fixed degrees of freedom in the chi-squared distribution may not fully capture the dynamic nature of financial return distributions, especially during periods of market stress. Secondly, the framework does not account for potential leverage effects or asymmetries in volatility, which may be better addressed using GJR-GARCH or EGARCH extensions. Thirdly, the application is limited to univariate time series analysis, whereas multivariate extensions could enhance risk evaluation in diversified portfolios. Additionally, the use of a single distribution family restricts the exploration of other heavy-tailed or skewed distributions that may offer superior tail risk modeling in specific contexts.

6. Concluding Remarks

The selection of an appropriate distribution is essential for accurately estimating the quantiles of financial return distributions. Financial returns often exhibit skewness, excess kurtosis, and fat tails, making standard normal distribution assumptions problematic. If an incorrect distribution is used, the estimated risk measures may underestimate or overestimate the actual risk, leading to either excessive capital reserves or insufficient risk coverage. In this work, we have derived the following:

$$\text{VaR}_p(X) = 2 \cdot Q^{-1}\left(\frac{m}{2}, 0, p\right), \quad \text{CVaR}_p(X) = \frac{2\Gamma\left(\frac{m}{2} + 1, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right)}{\Gamma\left(\frac{m}{2}, Q^{-1}\left(\frac{m}{2}, 0, p\right)\right)},$$

where $Q^{-1}(\cdot)$ denotes the inverse generalized regularized incomplete gamma function. These results are then operationalized in a risk management context by modeling time-varying volatility in stock returns using the GARCH process. The chi-squared distribution arises as a result of summing the squared values of m independent Gaussian random variables, each possessing a mean of zero and a unit variance. More generalized forms of this distribution can be derived by considering the sum of squares of Gaussian random variables with different statistical properties. An extension of this concept is the noncentral chi-squared distribution, which is attained when summing the squared values of independent Gaussian random variables that maintain a unit variance but have nonzero means. Exploring these extended distributions for the computation of risk measures, like VaR and CVaR, and analyzing their practical relevance in portfolio risk management within the framework of GARCH models presents directions for future research. However, empirical validation of this assumption through statistical goodness-of-fit tests—such as the Kolmogorov–Smirnov or Anderson–Darling tests—was not performed in the current version. Incorporating such tests to compare the chi-squared distribution against alternative models like the Student- t or generalized error distribution is a promising direction for future work to strengthen the empirical basis of the proposed methodology.

The simulation results presented in this study largely align with the theoretical expectations. The use of the chi-squared distribution allowed for capturing the skewness and excess kurtosis present in the financial return distributions, with minimal discrepancies in the VaR and CVaR measures when using confidence levels of 80% or 90%. These results

demonstrate that the chi-squared distribution, when combined with the GARCH model, provides a robust approach for risk quantification, particularly in volatile markets. However, slight deviations in the second experimental case suggest that further refinement, such as incorporating noncentral chi-squared distributions or more advanced GARCH models, could yield even more precise estimates for stocks with extreme volatility.

Author Contributions: Conceptualization, F.S.; Methodology, F.S.; Software, F.S.; Validation, F.S. and T.L.; Formal analysis, F.S.; Investigation, Q.M.; Resources, Q.M.; Data curation, Q.M.; Writing—original draft, Q.M.; Writing—review & editing, Q.M.; Visualization, T.L.; Supervision, T.L.; Project administration, T.L.; Funding acquisition, T.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Scientific Research Project of Jilin Provincial Department of Education (JJKH20251638KJ), the Open Fund Project of Marine Ecological Restoration and Smart Ocean Engineering Research Center of Hebei Province (HBMESO2321), the Technical Service Project of Eighth Geological Brigade of Hebei Bureau of Geology and Mineral Resources Exploration (KJ2022-021), and the Technical Service Project of Hebei Baodi Construction Engineering Co., Ltd. (KJ2024-012).

Data Availability Statement: No special data were utilized.

Conflicts of Interest: The authors declare no conflicts of interest.

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